Sai Kottamasu

469-980-1996 | saikottamasu123@gmail.com | linkedin.com/in/nagasaitejakottamasu | github.com/saikottamasu123-sys

Education

University of Texas at Dallas May 2028
B.S. in Computer Science, Minors: IT Systems & Business Analytics (GPA: 4.0 / 4.0), 3x Dean's List *Richardson, Texas*

- **Relevant Coursework:** Data Structures and Algorithms, Systems Programming in UNIX/C, Object Oriented Design in C++, Database Systems, Discrete Mathematics, Linear Algebra, Probability & Statistics

Experience

TLT (Tomorrow's Leaders Today) August 2026 – Current
Software Engineering Intern — Python, HTML/CSS, LangChain, LLMs *Richardson, Texas*

- Engineered an internal RAG assistant with LangChain and vector embeddings over organization documents, cutting recurring intern and staff support inquiries by 50%.
- Redesigned high-traffic website pages and refactored legacy HTML/CSS, improving estimated load times by 20–30% and delivering a faster, more consistent user experience.

ACM - Association of Computing Machinery August 2026 – Current
Full-Stack Developer — Node.js, React Native, Expo, FastAPI, PostgreSQL, Supabase, CI/CD *Richardson, Texas*

- Developed a full-stack campus marketplace in React Native and Expo, supporting student listings and buyer-seller messaging through Node.js and FastAPI services.
- Secured campus data with Supabase/PostgreSQL row-level security and Microsoft OAuth authentication.
- Built CI/CD workflows that automate integration checks and repeatable deployments, reducing release friction and keeping the React Native application deployable across team contributions.

Dallas Formula Racing January 2026 – September 2026
Software Engineer — Python, Apache Kafka, PostgreSQL, AWS S3, FastAPI, LangChain *Richardson, Texas*

- Built a Kafka pipeline streaming live vehicle sensor data into PostgreSQL for real-time telemetry analysis.
- Automated PostgreSQL backups to AWS S3 with pg_dump, enabling reliable telemetry database recovery.
- Developed a Python/LangChain RAG assistant for the DFR website that grounds responses in team content, helping users navigate resources and resolve common questions, reducing questions by 50%.

Research

RIDE Research – Fluid Dynamics Aug 2026 – Current
Undergraduate Researcher — Python, PyTorch, Scikit-learn, NumPy, Plotly *Richardson, Texas*

- Built a NumPy and Scikit-learn pipeline applying Proper Orthogonal Decomposition to compress high-dimensional PIV velocity fields into spatial modes and temporal coefficients.
- Trained a PyTorch CNN to predict missing POD coefficients from adjacent PIV snapshots.
- Reconstructed missing PIV velocity fields and validated predictions against original measurements with RMSE and spatial error analysis, visualizing results in Plotly.

Projects

Investly — *Python, LangGraph, LangChain, FastAPI, Google Gemini, ChromaDB, RESTful APIs*

- Engineered a multi-agent investment research system with LangGraph state machines coordinating eight parallel and sequential analysis nodes, cutting per-stock research time to under 40 seconds.
- Built a RAG pipeline over SEC EDGAR 10-K/10-Q filings with Google embeddings and ChromaDB, surfacing high-signal evidence across five investment-critical query dimensions.
- Delivered a FastAPI service combining live yfinance and EDGAR data into structured JSON with analyst-specific opinions and complete Markdown investment reports.

SightLine — *Python, FastAPI, Neo4j, React*

- Modeled supplier, customer, and competitor relationships across 500+ companies in Neo4j and served multi-hop traversal queries through a FastAPI intelligence API, adding indexes and Redis caching to cut p95 latency from 120 ms to 8 ms.
- Applied Neo4j Graph Data Science (PageRank, betweenness centrality, Louvain community detection) to identify critical suppliers and industry clusters, and built a disruption-propagation model estimating downstream exposure across 3 supplier tiers.
- Built a React and Cytoscape.js interface with relationship filtering, path highlighting, and risk overlays, backed by a data pipeline using entity resolution to ingest and deduplicate company relationships from public filings.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, C++, C, HTML/CSS
Frameworks & Tools: Node.js, React, React Native, FastAPI, RESTful APIs, Apache Kafka, AWS (S3), PostgreSQL, Neo4j, Docker, Git/GitHub, CI/CD, LangChain, LangGraph, ChromaDB, PyTorch, Scikit-learn, Pandas, NumPy, Plotly
Concepts: API Design, Event Streaming, Cloud Computing, Agile Methodology, Version Control, OOP, RAG, LLM Agents